IntelliCredit

GENERALIZED CREDIT ANALYSIS FRAMEWORK

+ Bug Fix Guide for Current UI Issues

What this document covers:

1. 5 Critical Bugs identified in your current IntelliCredit UI (with fixes)
2. Correct formulas for every metric your system computes
3. Generalized PDF extraction rules for any financial statement
4. NBFC vs Corporate logic — what to compute vs skip
5. Credit scoring engine with complete benchmark tables

Reference Company: Vivriti Capital Limited | FY 2024-25
Date: March 21, 2026

■ PART 1 — Bug Analysis: What Your IntelliCredit UI Is Getting Wrong

Based on the 4 screenshots provided, your IntelliCredit system has 5 critical calculation errors for Vivriti Capital Limited FY 2024-25. Each bug is analyzed below with the root cause, correct formula, and code-level fix.

Quick Summary — 5 Bugs at a Glance

#	Metric	System Shows	Correct Value	Error Type
1	Total Revenue / Sales	■64.46 Cr	■1,347 Cr	Wrong field extracted from PDF
2	Net Profit Margin	341.3%	16.13%	Wrong denominator — caused by Bug #1
3	Loan Book / AUM	■241.01 Cr	■9,081 Cr	Wrong field extracted from PDF
4	ROE %	2.82%	11.95%	ROE = ROA (same formula used for both)
5	Operating Cash Flow / DSCR	■1,07,640 Cr / 14.41x	■(1,076) Cr / N/A	Unit conversion error — lakhs shown as crores

Note: ICR shows 4.00 on the Analysis dashboard but N/A on the Credit Ratios table. This inconsistency also needs to be resolved — it should show N/A consistently since EBIT is not available for an NBFC (finance costs are operating costs, not leverage costs).

BUG #1 — Revenue Shows ■64.46 Cr Instead of ■1,347 Cr

System shows →

```
Revenue / Sales = ■64.46 Cr (shown in Analysis dashboard and Step 2 table)
```

Why it's wrong →

■64.46 Cr is the Fee & Commission Income only (INR 6,543.76 lakhs = ■65.4 Cr). The system is extracting the fee income line from the P&L; instead of Total Revenue from Operations.

Root cause →

PDF parser is matching keyword 'Revenue' or 'Income' and picking the FIRST matching line in the P&L;, which is often a sub-component (fees, commissions) rather than the top-line total. In Vivriti's P&L;: Interest Income = ■1,270 Cr, Fee Income = ■65.4 Cr, Total = ■1,347 Cr.

Correct formula →

```
Total Revenue = Interest Income + Fee & Commission Income + Net Gain on Fair Value + Other Income = 1,26,996.57 + 6,543.76 + 155.59 + 1,015.26 + 1,684.65 (in lakhs) = 1,36,395.83 lakhs = ■1,364 Cr OR simply: look for 'Total income' or 'Total revenue from operations' line in P&L;
```

Fix in code →

```
# WRONG — extracts first 'income' line: revenue = extract_field(pdf, 'Fee and commission income') # CORRECT — extract the TOTAL line: revenue = extract_field(pdf, 'Total income') or revenue = extract_field(pdf, 'Total revenue from operations') # Always prefer the AGGREGATED total, not a sub-component
```

METRIC	WRONG (current)	CORRECT (should be)
Total Revenue	**■64.46 Cr**	**■1,347 Cr**

	Fee income sub-component	*Total Revenue from Operations + Other Income*

BUG #2 **PAT Margin Shows 341.3% — Caused by Bug #1**

System shows →	`Net Profit Margin = 341.3% (■220 Cr PAT / ■64.46 Cr revenue)`
Why it's wrong →	Net Profit Margin cannot exceed 100% for a normal business. 341.3% means PAT > Revenue which is impossible in a P&L; context. This is a direct downstream consequence of Bug #1 — wrong revenue denominator.
Root cause →	Bug #1 extracted fee income (■64.46 Cr) as Revenue. The margin formula PAT/Revenue then becomes 220/64.46 = 341.3%. With correct revenue (■1,364 Cr): 220/1364 = 16.13% — which matches the stated 16.13% in the AR.
Correct formula →	`Net Profit Margin = PAT / Total Revenue × 100 = 220 / 1,364 × 100 = 16.13%`
Fix in code →	`# Downstream fix — once Bug #1 is fixed, this auto-corrects: revenue = get_total_revenue(pdf) # Fix Bug #1 first net_profit_margin = (pat / revenue) * 100 # Add sanity check: if net_profit_margin > 100: raise ValueError('Margin > 100% — check revenue extraction')`

METRIC	WRONG (current)	CORRECT (should be)
Net Profit Margin	**341.3%**	**16.13%**
	PAT ÷ Fee Income (wrong)	*PAT ÷ Total Revenue (correct)*

BUG #3	Loan Book / AUM Shows ■241.01 Cr Instead of ■9,081 Cr
System shows →	`Loan Book (AUM) = ■241.01 Cr (shown in NBFC-Specific Metrics section)`
Why it's wrong →	■241.01 Cr is nowhere close to the actual AUM of ■9,081 Cr stated in MD&A.; ■241 Cr appears to be a specific loan sub-category (possibly residential mortgages = ■33,692 lakhs? or a single Note-level figure) being picked instead of the total Net Loan Book.
Root cause →	PDF parser is looking for 'Loan' keyword in Notes to Accounts and picking a specific loan category (a line in the loan breakdown table) instead of the TOTAL Net Loans figure. The Note on Loans (Note 7) in Vivriti's AR shows multiple sub-rows — system is picking one row instead of the 'Total Loans = (A-B)' line.
Correct formula →	`AUM = Stated AUM from MD&A; Key Highlights section = ■9,081 Cr OR: Net Loans = Total Loans (A) - Impairment Allowance (B) = from Balance Sheet 'Loans' line = ■8,64,668.56 lakhs = ■8,647 Cr (on-balance-sheet) Always prefer MD&A; stated AUM as it includes off-balance-sheet items`
Fix in code →	`# WRONG — picking a sub-row from the loan notes table: loan_book = extract_note_row(pdf, 'Loans', row_index=3) # CORRECT — use the final net total line or MD&A; stated value: loan_book = (extract_mda_value(pdf, 'AUM') or extract_balance_sheet_line(pdf, 'Loans') # The single Loans line on BS) # The Balance Sheet has ONE 'Loans' line = net of provisions = correct value`

METRIC	WRONG (current)	CORRECT (should be)
Loan Book / AUM	**■241.01 Cr**	**■9,081 Cr**
	Specific loan sub-category from Notes	*Total AUM from MD&A; or Net Loans from Balance Sheet*

BUG #4	ROE Shows 2.82% — Same as ROA (Wrong Formula)
System shows →	`Return on Equity (ROE) % = 2.82% (identical to ROA = 2.82%)`
Why it's wrong →	ROA and ROE cannot be identical for a leveraged company like Vivriti (D/E = 3.48x). When a company uses debt, ROE is always higher than ROA due to financial leverage. The correct ROE is 11.95% (stated in MD&A;). ROA = 2.82% is correct.
Root cause →	The system is using the same formula for both ROA and ROE — likely: ROE = PAT / Total Assets (which is the ROA formula). The denominator should be Shareholders' Equity (Net Worth), not Total Assets.
Correct formula →	`ROA = PAT / Average Total Assets × 100 = 220 / ((10,468 + 9,189) / 2) × 100 = 220 / 9,829 × 100 = 2.24% (or use stated 2.82%) ROE = PAT / Average Shareholders Equity × 100 = 220 / ((2,147 + 1,894) / 2) × 100 = 220 / 2,021 × 100 = 10.9% (stated: 11.95%) Always use AVERAGE (opening + closing) / 2 for both denominators`
Fix in code →	`# WRONG — same formula for both: roa = pat / total_assets * 100 roe = pat / total_assets * 100 # BUG: should be equity # CORRECT: avg_assets = (total_assets_current + total_assets_prior) / 2 avg_equity = (equity_current + equity_prior) / 2 roa = (pat / avg_assets) * 100 roe = (pat / avg_equity) * 100 # avg_equity = avg Net Worth = Equity Share Cap + Preference Cap + Other Equity`

METRIC	WRONG (current)	CORRECT (should be)
ROE %	**2.82%**	**11.95%**
	PAT ÷ Total Assets (ROA formula)	*PAT ÷ Avg. Shareholders Equity*

BUG #5	Unit Conversion Error — Lakhs Shown as Crores (CFO and Interest Expense)
System shows →	`Operating Cash Flow = ■1,07,640.69 Cr | Interest Expense = ■1,20,044.91 Cr`
Why it's wrong →	These numbers are actually in LAKHS (as stated in the financial statements header). ■1,07,640.69 lakhs = ■1,076 Cr (CFO). ■1,20,044.91 lakhs = ■1,200 Cr (interest). The system is reading the number but not dividing by 100 to convert to crores. This also makes DSCR = 1,07,640 / (1,20,044/12) = nonsensical 10-14x.
Root cause →	Vivriti's financial statements are stated in 'Rupees lakhs' (1 lakh = ■1,00,000). 1 Crore = 100 Lakhs. So to convert: Crores = Lakhs / 100. The system is not reading the unit header from the financial statement and not applying conversion.
Correct formula →	`Operating Cash Flow (CFO) = -1,07,640.69 lakhs ÷ 100 = ■(1,076.4) Cr [NEGATIVE — outflow] Interest Expense = 69,926.89 lakhs ÷ 100 = ■699.3 Cr (from Finance Costs in P&L;) DSCR for NBFC = N/A (Operating Cash Flow is structurally negative for growing NBFCs) DSCR is only meaningful for manufacturing/corporate borrowers`
Fix in code →	`# Step 1: Always read unit header from the financial statement unit_header = extract_header(pdf, 'amounts are in') # 'lakhs' or 'crores' divisor = 100 if 'lakhs' in unit_header.lower() else 1 # Step 2: Apply divisor to all extracted numbers cfo = extract_cashflow(pdf, 'Net Cash from Operating') / divisor interest = extract_pl(pdf, 'Finance costs') / divisor # Step 3: For NBFC, mark DSCR as N/A dscr = 'N/A — Not applicable for NBFC (CFO is structurally negative)' if company_type == 'NBFC': dscr = 'N/A'`

METRIC	WRONG (current)	CORRECT (should be)
Operating Cash Flow	**■1,07,640 Cr**	**■(1,076) Cr**
	Lakhs value treated as Crores	*Value ÷ 100 = correct Crore figure*

METRIC	WRONG (current)	CORRECT (should be)
DSCR	**14.41x**	**N/A**
	Computed on wrong CFO (unit error)	*Not applicable for NBFC*

■ PART 2 — Complete Corrected Output for Vivriti Capital FY 2024-25

Below is the full corrected set of values your system should output for Vivriti Capital. Use this as your ground truth for testing and validation.

Step 2 — Core Financial Inputs (Corrected)

METRIC	WRONG (Your UI)	CORRECT VALUE	SOURCE IN PDF
Total Revenue / Sales	**■64.46 Cr**	**■1,364 Cr**	P&L;: 'Total income' = 1,36,395.83 lakhs
EBITDA	**N/A ✓**	**N/A ✓**	Correct — not applicable for NBFC
EBIT	**N/A ✓**	■290.6 Cr (PBT proxy)	P&L;: 'Profit before tax' = 29,063.38 lakhs
Net Profit / PAT	**■220 Cr ✓**	**■220.04 Cr ✓**	P&L;: 'Net profit after tax' = 22,004.08 lakhs
Total Debt	**■7,470 Cr ✓**	**■7,471 Cr ✓**	MD&A; / Key Highlights section
Net Worth	**■2,146.91 Cr ✓**	**■2,146.91 Cr ✓**	Balance Sheet: Total Equity = 2,14,691.43 lakhs
Current Assets	**N/A ✓**	**N/A — NBFC ✓**	Correct — not applicable for NBFC
Current Liabilities	**N/A ✓**	**N/A — NBFC ✓**	Correct — not applicable for NBFC
Operating Cash Flow (CFO)	**■1,07,640 Cr ✗**	**■(1,076) Cr**	Cash Flow Stmt: -1,07,640.69 lakhs ÷ 100
Interest Expense (Finance Costs)	**■1,20,044 Cr ✗**	**■699.3 Cr**	P&L;: Finance costs = 69,926.89 lakhs ÷ 100

NBFC-Specific Metrics (Corrected)

Metric	Your UI Shows	Correct Value	Formula / Source
Loan Book / AUM	■241.01 Cr ✗	■9,081 Cr	MD&A; Key Highlights: 'AUM increased to INR 9,081.07 Cr'
Net Interest Income (NII)	Not Available	■570.7 Cr (calculated)	Interest Income – Finance Costs = 12,699.66 – 6,992.69 (Cr) = 570.7 Cr
Net Interest Margin (NIM)	Not Available	~7.16% (estimated)	NII ÷ Avg Loan Book = 570.7 ÷ ((9,081+7,830)/2) = 570.7/8,456
Gross NPA %	1.89% ✓	1.89% ✓	Directly stated in MD&A; / Key Financial Ratios
Net NPA %	0.71% ✓	0.71% ✓	Directly stated in MD&A; / NPA Movement Note
Capital Adequacy Ratio (CRAR)	21.02% ✓	21.02% ✓	Directly stated in Notes on CRAR (Note 47)

Cost of Funds	Not Available	~9.35% (estimated)	Finance Costs ÷ Avg Borrowings = 699.3 ÷ ((7,471+6,569)/2) = 699.3/7,020
Return on Assets (ROA)	2.82% ✓	2.82% ✓	Stated in MD&A.; Also: PAT/Avg Assets = 220/7,829 = 2.81%
Return on Equity (ROE)	2.82% ✗	11.95%	PAT ÷ Avg Equity = 220 ÷ ((2,147+1,894)/2) = 220/2,021 = 10.9% (stated: 11.95%)

Step 3 — Computed Credit Ratios (Corrected)

Ratio	Formula	Your UI	Correct	Interpretation
D/E Ratio	Total Debt / Net Worth	3.48x ✓	3.48x ✓	Correct. Normal for NBFC (RBI limit ~7x).
ICR	EBIT / Finance Costs	4.00 / N/A ✗	N/A for NBFC	Finance costs ARE operating costs for NBFC. ICR is not meaningful. Mark N/A.
DSCR	Op.CF / Debt Service	14.41x ✗	N/A for NBFC	CFO is structurally negative for growing NBFC. DSCR is N/A.
Current Ratio	Curr. Assets / Curr. Liab.	N/A ✓	N/A ✓	Correct. Not applicable for NBFC.
EBITDA Margin	EBITDA / Revenue × 100	N/A ✓	N/A ✓	Correct. Not applicable for NBFC.
Net Profit Margin	PAT / Revenue × 100	341.3% ✗	16.13%	220 ÷ 1,364 = 16.1%. Matches stated 16.13% in Key Ratios.
ROA	PAT / Avg. Assets × 100	2.82% ✓	2.82% ✓	Correct (matches stated value).
ROE	PAT / Avg. Equity × 100	2.82% ✗	11.95%	11.95% stated in MD&A.; Leverage amplifies ROE above ROA.
NIM (add this)	NII / Avg. Loan Book × 100	Missing	~7.16%	Key NBFC metric. Should be added to your ratio table.
Credit Cost (add)	Impairment / Avg. Loans × 100	Missing	~2.1%	19,380 lakhs ÷ avg 8,066 Cr = 2.4%. Rising trend — risk signal.

■ PART 3 — Generalized Framework: How Your Code Should Work for ANY PDF

This section defines the complete generalized algorithm. Follow this exact sequence in your code for any uploaded financial statement.

Step A — Classify the Company Type (Do This First)

Before extracting any number, your system must determine company type. This decides WHICH metrics to compute and which to mark N/A.

Look for this in the PDF	→ Classify as	→ Metrics to use
'NBFC', 'Non-Banking Financial Company', 'RBI registration', 'CRAR', 'NPA', 'Certificate of Registration under RBI Act'	NBFC / HFC	NII, NIM, GNPA, NNPA, PCR, CRAR, AUM, Credit Cost, Cost of Funds. Skip: EBITDA, Inventory Days, Debtors Days, Current Ratio, DSCR, ICR.
'Scheduled Commercial Bank', 'CASA Ratio', 'SLR', 'CRR', 'Banking License'	Bank	Same as NBFC + CASA, CD Ratio, LCR. Skip manufacturing metrics.
Inventory in Balance Sheet, COGS in P&L;, no RBI registration	Manufacturing / Corporate	EBITDA, ICR, DSCR, Inventory Days, Debtors Days, Creditors Days. Skip: NPA, CRAR, NIM, AUM.
'Software', 'SaaS', 'IT Services', high intangibles, low tangibles	Technology	Revenue growth, EBITDA margin, Free Cash Flow. DSCR if project financed.
'Concession Agreement', 'Annuity', 'Infrastructure Project'	Infrastructure	DSCR, LLCR, Debt tenor vs asset life. Sponsor quality critical.

Step B — Detect Financial Statement Units (Critical for Bug #5 Fix)

Always read the unit declaration before extracting any number.

```
# Pattern 1: 'All amounts are in Rupees lakhs' # Pattern 2: 'Rs. in Crore' / '(Rs. Crores)'
# Pattern 3: 'INR in Millions' unit_text = extract_text_near_financial_table(pdf) if
'lakhs' in unit_text.lower() or 'lakh' in unit_text.lower(): multiplier = 0.01 # divide by
100 to convert lakhs → crores elif 'crore' in unit_text.lower(): multiplier = 1 # already
in crores elif 'million' in unit_text.lower(): multiplier = 0.1 # 10 million = 1 crore #
Apply multiplier to EVERY extracted number: revenue = raw_revenue_from_pdf * multiplier
```

★ *Vivriti Capital's statements are in 'Rupees lakhs'. Tata Capital's statements are in 'Rs. in crore'. Always detect — never hardcode the unit.*

Step C — Exact Extraction Map: Where to Find Each Metric in Any Annual Report PDF

Metric	Primary Location	Search Keywords	Fallback / Notes
Total Revenue	Statement of Profit & Loss — last line of Revenue section	'Total income', 'Total revenue from operations'	NEVER use a sub-component like fee income or interest income alone
Interest Income	P&L; — Revenue from Operations breakdown, first line	'Interest income', 'Finance income'	For NBFC this is usually 85-95% of total revenue
Finance Costs	P&L; — Expenses section	'Finance costs', 'Interest expense', 'Borrowing costs'	For NBFC this is the primary operating cost
PAT	P&L; — bottom section	'Net profit after tax', 'Profit for the year', 'PAT'	For consolidated: use 'attributable to owners of company' not total
PBT / EBIT proxy	P&L; — above tax line	'Profit before tax', 'PBT'	For NBFC use PBT as EBIT proxy; for corporate: PBT + Finance Costs = EBIT
Total Assets	Balance Sheet — final total line on Assets side	'Total assets', 'Total Assets'	Must equal Total Liabilities + Total Equity
Net Worth / Equity	Balance Sheet — Equity section total	'Total equity', 'Equity attributable to owners'	= Equity Share Cap + Preference Cap + Other Equity (Reserves)
Total Borrowings	Balance Sheet — Financial Liabilities + Notes on Borrowings	'Debt securities', 'Borrowings', 'NCDs', 'Subordinated liabilities'	Sum: Debt Securities + Borrowings other than debt securities + Sub-debt
Loan Book / AUM	FIRST CHOICE: MD&A; Key Highlights ('AUM increased to X'). SECOND: Balance Sheet 'Loans' line	'AUM', 'Loan book', 'Net loans', 'Total loans'	Balance Sheet Loans line = single line = Net of ECL provisions = correct
GNPA %	MD&A; highlights table OR Key Financial Ratios section OR NPA Movement note	'Gross NPA', 'GNPA', 'Non-Performing Assets'	Usually stated as percentage. Cross-check with NPA movement note
NNPA %	Same as GNPA — usually in same table	'Net NPA', 'NNPA', 'Net Non-Performing'	If not stated: (Gross NPA - Provisions) / Net Loan Book × 100
PCR	MD&A; or GNPA table	'Provision coverage ratio', 'PCR'	If not stated: (Gross NPA - Net NPA) / Gross NPA × 100
CRAR	Board's Report OR Notes to Accounts (Note ~47 in NBFC ARs)	'CRAR', 'Capital adequacy ratio', 'Capital to risk'	ALWAYS use stated value — never calculate. Stated by management.
ROA / ROE	MD&A; highlights or Key Financial Ratios table	'Return on assets', 'Return on equity', 'RoA', 'RoE'	Use stated if available. Calculate only as fallback (see formulas below).
D/E Ratio	Board's Report — Key Financial Ratios OR MDA	'Debt to equity', 'D/E ratio', 'Gearing'	If not stated: Total Debt / Net Worth

Cash Flow from Operations	Statement of Cash Flows — Section A	'Net cash from operating activities', 'Net cash used in operating'	Will be NEGATIVE for growing NBFCs — this is NORMAL
EBITDA	NOT usually stated — must calculate from P&L;	No direct keyword — derive from PBT	EBITDA = PBT + Finance Costs + Depreciation. SKIP for NBFC.
Inventory	Balance Sheet — Current Assets section	'Inventories', 'Stock in trade', 'Raw materials'	If not present → company is not manufacturing. Mark N/A.
Trade Receivables	Balance Sheet — Current Assets / Financial Assets	'Trade receivables', 'Debtors'	For NBFC: ignore. Loans are the receivables — use NPA instead.
Promoter Holding %	Shareholding Pattern table (listed). Board composition for unlisted.	'Promoter', '% of holding', '% holding'	For unlisted VC-backed: look at Nominee Director table for investor names
Credit Rating	Board's Report — Finance or Credit Rating subsection	'CRISIL', 'ICRA', 'CARE', 'India Ratings', 'S&P;', 'Fitch'	State: Agency + Rating + Outlook e.g. 'CRISIL AAA/Stable'

Step D — Complete Formula Reference: How to Calculate Each Metric

Apply these formulas in your code after extracting the raw numbers.

Revenue YoY Growth %

```
(Revenue_FY25 - Revenue_FY24) / Revenue_FY24 × 100
```

Flag if >50% — check for M&A; or merger impact (like TMFL). Use 'ex-merger' figures if available.

EBITDA [Corporate only]

```
EBITDA = PBT + Finance Costs + Depreciation & Amortisation
```

Do NOT compute for NBFC. Mark as 'N/A — Not applicable for NBFC'.

EBITDA Margin % [Corporate]

```
EBITDA Margin = EBITDA / Total Revenue × 100
```

Benchmark: >20% = strong mfg; >30% = strong tech/pharma. Skip for NBFC.

PAT Margin %

```
PAT Margin = PAT / Total Revenue × 100
```

Sanity check: must be between -200% and 100%. If outside range, check revenue extraction.

ROA %

```
ROA = PAT / ((Total Assets_FY25 + Total Assets_FY24) / 2) × 100
```

Use AVERAGE assets. If prior year not available: use current year assets. Benchmark: NBFC >2%; Corporate >1%.

ROE %

```
ROE = PAT / ((Net Worth_FY25 + Net Worth_FY24) / 2) × 100
```

Net Worth = Equity Share Cap + Pref Cap + Other Equity. ROE > ROA always if D/E > 0. If ROE = ROA, check your denominator.

D/E Ratio

```
D/E = Total Debt / Net Worth Total Debt = Debt Securities + Borrowings + Subordinated Liabilities
```

NBFC acceptable: 3-7x. Corporate acceptable: 1-3x. Use stated value from Key Ratios if available.

NII [NBFC only]

```
NII = Interest Income - Finance Costs
```

Some companies report 'NII including fee income' in MD&A; — note which definition is used.

NIM % [NBFC only]

```
NIM = NII / ((Loan Book_FY25 + Loan Book_FY24) / 2) × 100
```

Use net loan book (after ECL provisions) for denominator. Benchmark: >6% = strong.

Cost of Funds % [NBFC only]

```
CoF = Finance Costs / ((Total Borrowings_FY25 + Total Borrowings_FY24) / 2) × 100
```

Compare to lending yield (Interest Income / Avg. Loan Book) to compute spread. Spread > 3% is healthy.

Credit Cost % [NBFC only]

```
Credit Cost = Impairment on Financial Instruments (P&L;) / ((Loan Book_FY25 + Loan Book_FY24)/2) × 100
```

Benchmark: <1% = excellent; 1-2% = normal; >2% = elevated; >3% = stress.

PCR % [NBFC only]

```
PCR = (Gross NPA - Net NPA) / Gross NPA × 100 OR PCR = Total Provisions for NPA / Gross NPA × 100
```

Both methods should give same result. If stated directly, use stated. Benchmark: >70%.

ICR [Corporate only]

```
ICR = EBIT / Finance Costs OR ICR = EBITDA / Finance Costs (more conservative)
```

For NBFC: Mark as N/A. Finance costs are operating costs — ICR is not meaningful. Benchmark for corporate: >2.5x.

DSCR [Corporate only]

```
DSCR = (PAT + Depreciation + Finance Costs) / (Annual Principal Repayment + Annual Interest) = Net Cash Available for Debt Service / Total Debt Service
```

For NBFC: Mark as N/A (CFO is structurally negative). Principal repayment from maturity schedule in Notes on Borrowings. Benchmark: >1.5x.

Debtors Days [Corporate only]

```
Debtors Days = (Trade Receivables / Revenue) × 365
```

For NBFC: N/A. Benchmark for corporate: <45 days = good; >90 days = concern.

Inventory Days [Manufacturing only]

```
Inventory Days = (Inventory / COGS) × 365
```

Only if Inventory exists in Balance Sheet. COGS from P&L; cost breakdown.

Current Ratio [Corporate only]

```
Current Ratio = Current Assets / Current Liabilities
```

For NBFC: N/A. Corporate benchmark: >1.2x.

Working Capital [Corporate only]

```
Working Capital = Current Assets - Current Liabilities
```

For NBFC: N/A. For corporate: should be positive.

Tangible Net Worth

```
Tangible NW = Net Worth - Intangible Assets - Goodwill
```

Used in lender covenants. Intangibles from Notes on Other Intangible Assets.

■ PART 4 — Credit Scoring Engine: Complete Algorithm

This is the exact scoring algorithm your system should implement. Total score = 100 points across 4 categories. Scores feed into a grade and lending recommendation.

Category A — Capital & Solvency (25 Points)

Sub-Parameter	Max	How to Score (NBFC)	How to Score (Corporate)
CRAR / Capital Adequacy	10 pts	≥20% → 10 | 17-20% → 8 | 15-17% → 6 | <15% → 0 (breach)	N/A for corporate — use D/E scoring instead
Promoter / Sponsor Quality	10 pts	Listed group / sovereign → 9-10 | Strong PE/VC → 7-8 | Founder-only → 6 | Unclear → 4	Same rule applies for all company types
Debt-to-Equity Ratio	5 pts	NBFC: <3x → 5 | 3-5x → 4 | 5-7x → 3 | >7x → 1	Corporate: <1x → 5 | 1-2x → 4 | 2-3x → 3 | 3-4x → 2 | >4x → 1

Category B — Asset Quality (25 Points)

Sub-Parameter	Max	How to Score (NBFC)	How to Score (Corporate)
GNPA % / Default History	10 pts	<1% → 10 | 1-1.5% → 8 | 1.5-2% → 7 | 2-3% → 5 | 3-5% → 3 | >5% → 1	No defaults → 10 | Delayed payments → 7 | Restructured → 5 | NPA → 2
NNPA % / Receivables Quality	10 pts	<0.5% → 10 | 0.5-1% → 8 | 1-2% → 6 | 2-3% → 4 | >3% → 2	Debtors Days <45d → 9 | 45-75d → 7 | 75-120d → 5 | >120d → 3
PCR / Provisioning Buffer	5 pts	>75% → 5 | 65-75% → 4 | 60-65% → 3 | 50-60% → 2 | <50% → 1	Strong contingency reserve → 5 | Minimal → 3 | None → 1

★ *TREND PENALTY: If NPA additions grew >50% YoY, deduct 1-2 pts from Category B total. A worsening trend is more dangerous than a high but stable level.*

Category C — Profitability & Returns (25 Points)

Sub-Parameter	Max	How to Score (NBFC)	How to Score (Corporate)
ROA %	10 pts	>3% → 10 | 2-3% → 8 | 1.5-2% → 6 | 1-1.5% → 5 | <1% → 3	>3% → 10 | 2-3% → 8 | 1-2% → 6 | 0.5-1% → 4 | <0.5% → 2
ROE %	10 pts	>18% → 10 | 14-18% → 8 | 10-14% → 7 | 6-10% → 5 | <6% → 3	>20% → 10 | 15-20% → 8 | 10-15% → 6 | 5-10% → 4 | <5% → 2
PAT Growth % (YoY)	5 pts	>25% → 5 | 15-25% → 4 | 5-15% → 3 | 0-5% → 2 | Negative → 1	Same. Adjust for one-time exceptional items before scoring.

Category D — Business Quality & Outlook (25 Points)

Sub-Parameter	Max	Scoring Rule (Same for All Company Types)
Credit Rating	10 pts	AAA / AA+ → 10 | AA / AA- → 9 | A+ / A → 7.5 | A- / BBB+ → 6 | BBB → 4.5 | BB or lower → 2 | Unrated → 4

Scale & Diversification	5 pts	Market leader / large / multi-product / multi-geography → 5 | Mid-scale / niche → 3.5 | Small / single product → 2
Management & Governance	5 pts	Clean audit, experienced board, no penalties → 5 | Minor issues → 3.5 | Audit qualification / RBI penalty → 2 | Adverse opinion → 0
Funding Access & Outlook	5 pts	Multiple channels, international access, IPO/expansion → 5 | Good domestic access → 3.5 | Limited / concentrated → 2

Grade Mapping — Score → Credit Grade → Lending Decision

Score Range	Internal Grade	Equivalent External	Lending Decision
90 – 100	AAA	CRISIL AAA / ICRA AAA	Approve — best pricing, maximum limit
85 – 89	AA+	CRISIL AA+ / ICRA AA+	Approve — standard pricing
80 – 84	AA	CRISIL AA / ICRA AA	Approve — normal covenants
75 – 79	AA-	CRISIL AA- / ICRA AA-	Approve — mild monitoring covenants
70 – 74	A+/BBB+	CRISIL A+ / ICRA A+	Approve — periodic review required
65 – 69	BBB	CRISIL BBB / ICRA BBB	Conditional — tighter covenants, collateral
55 – 64	BB+	CRISIL BB+ / ICRA BB+	Decline or require full collateral + guarantee
45 – 54	BB	CRISIL BB / ICRA BB	Decline — return for resubmission only
Below 45	B/CCC	Sub-investment grade	Decline / exit if existing exposure

Auto Red Flag Rules — Always Flag Regardless of Score

- ■ GNPA > 4% for NBFC OR Net NPA > 2% for NBFC
- ■ PCR declining by > 10 percentage points in a single year
- ■ NPA additions > 2× prior year additions (rapid worsening trend)
- ■ CRAR below regulatory minimum (15% for NBFC, 11.5% for bank)
- ■ Negative Net Worth (Total Liabilities exceed Total Assets)
- ■ ICR < 1.0x for manufacturing company (interest not covered by operations)
- ■ DSCR < 1.0x for corporate borrower (debt service not covered)
- ■ Qualified / adverse / disclaimer audit opinion
- ■ RBI penalty or regulatory action in Board's Report
- ■ PAT Margin > 100% or < -100% → data extraction error in your system
- ■ ROE = ROA exactly → same formula used for both (implementation bug)
- ■ Any metric value that implies division by zero or unit mismatch
- ■ Co-founder reclassification from Promoter to Non-Promoter category

- Major associate / subsidiary loss that may require parent capital support

■ PART 5 — Quick Reference Cheat Sheet

Metric	Formula	NBFC?	Corp?	Good	Warn	Red Flag
Total Revenue	Sum of all P&L; income lines	✓	✓	Growing	Flat	Falling
PAT Margin	PAT / Revenue × 100	✓	✓	>15%	5-15%	<5%
ROA	PAT / Avg.Assets × 100	✓	✓	>2%	1-2%	<1%
ROE	PAT / Avg.Equity × 100	✓	✓	>15%	8-15%	<8%
D/E	Total Debt / Net Worth	✓	✓	NBFC<5x / Corp<2x	NBFC 5-7x	>7x/4x
NII	Int.Income - Finance Costs	✓	N/A	Growing	Flat	Falling
NIM %	NII / Avg.Loan Book × 100	✓	N/A	>6%	4-6%	<4%
AUM / Loan Book	MD&A; stated or BS Loans line	✓	N/A	Growing	Flat	Declining
GNPA %	Gross NPA / Gross Loans × 100	✓	N/A	<1.5%	1.5-3%	>3%
NNPA %	Net NPA / Net Loans × 100	✓	N/A	<0.5%	0.5-1.5%	>1.5%
PCR %	(GNPA-NNPA)/GNPA × 100	✓	N/A	>70%	60-70%	<60%
CRAR %	Stated in AR notes	✓	N/A	>18%	15-18%	<15%
Cost of Funds	Finance Costs / Avg.Borr × 100	✓	N/A	Low vs yield	Rising	>Yield
Credit Cost	Impairment / Avg.Loans × 100	✓	N/A	<1%	1-2%	>2%
EBITDA Margin	EBITDA / Revenue × 100	N/A	✓	>20%	10-20%	<10%
ICR	EBIT / Finance Costs	N/A	✓	>3x	1.5-3x	<1.5x
DSCR	NCADS / Debt Service	N/A	✓	>1.5x	1.25-1.5x	<1.25x
Current Ratio	Curr.Assets / Curr.Liab	N/A	✓	>1.5x	1-1.5x	<1x
Debtors Days	Receivables/Revenue × 365	N/A	✓	<45d	45-90d	>90d
Inventory Days	Inventory/COGS × 365	N/A	Mfg	<60d	60-120d	>120d
CFO	From Cash Flow Statement	✓	✓	NBFC: N/A Corp:>0	N/A	Corp: <0 for 3yrs

★ *Sanity checks your code must always run: (1) PAT Margin must be between -200% and 100%. (2) ROE must be ≥ ROA for profitable companies with positive debt. (3) GNPA ≥ NNPA always. (4) Total Assets = Total Liabilities + Total Equity (balance check). (5) If any extracted figure is >1000 times the expected order of magnitude, suspect a unit conversion error.*